



Linear Independence

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Introduction



Price Problem



\$ 70'000



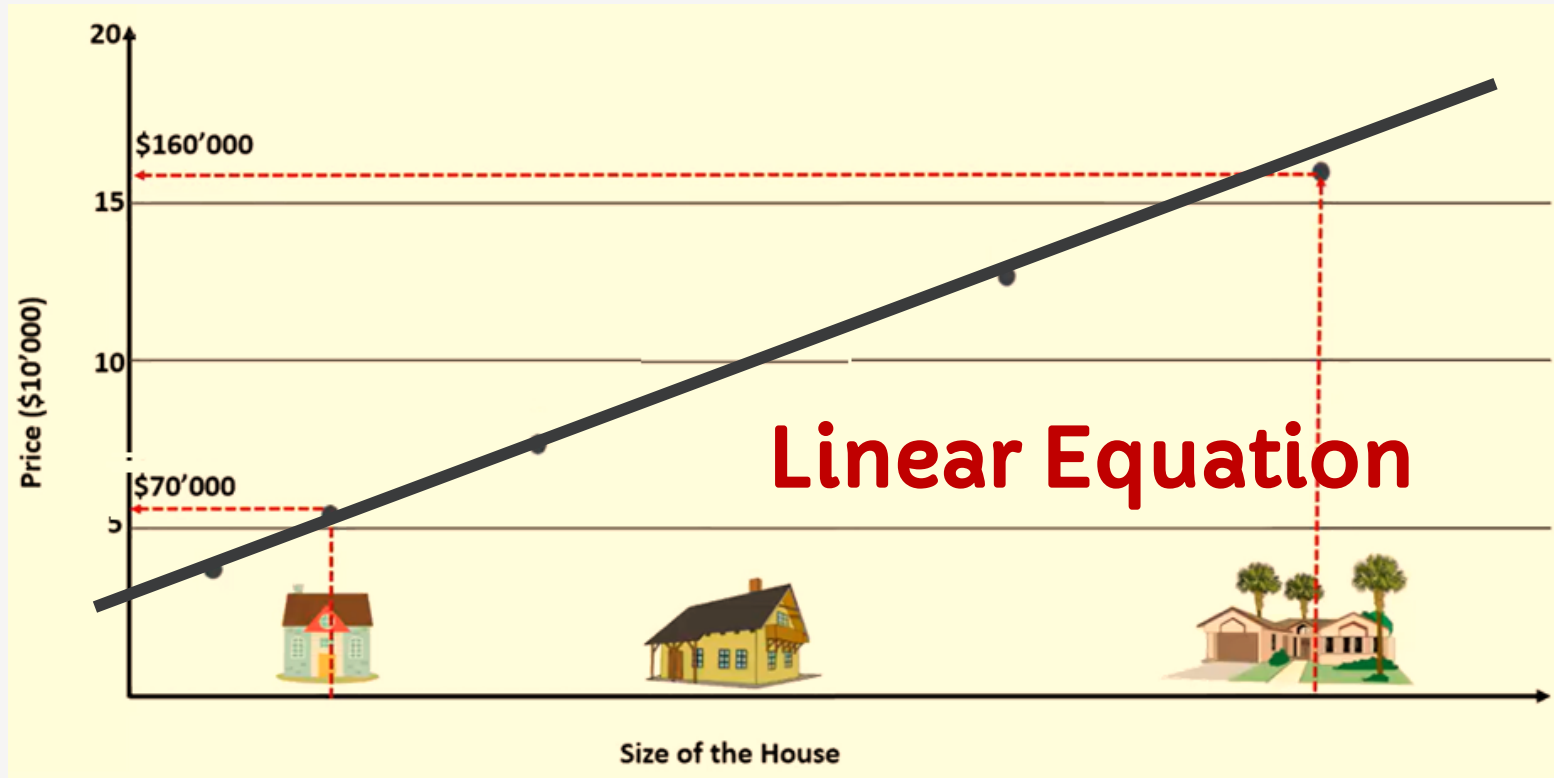
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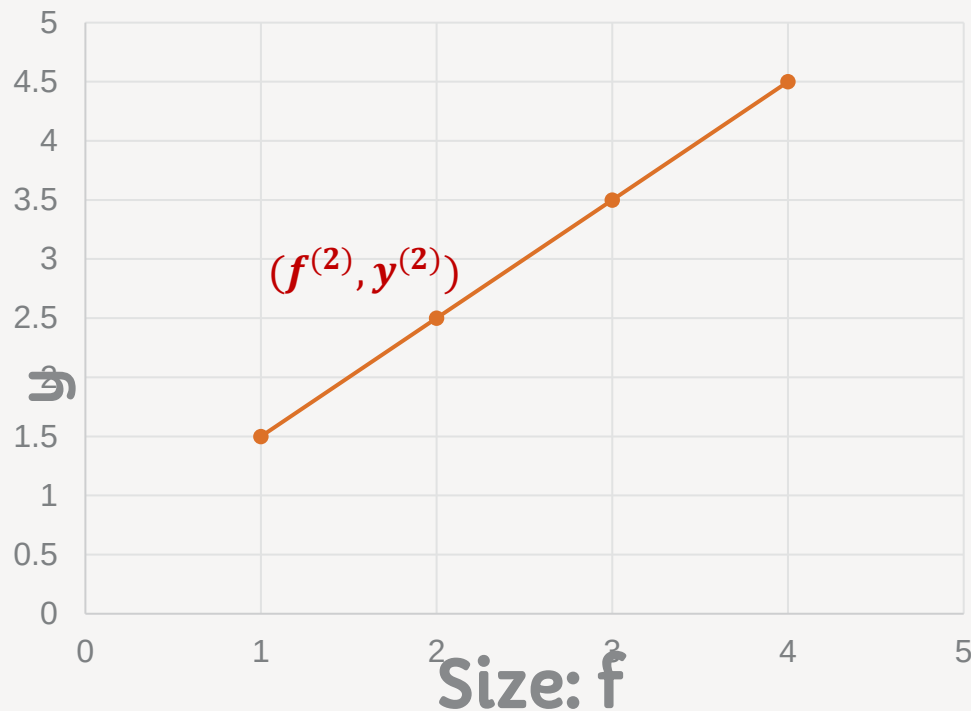
\$ 160'000



Price Problem



Linear Equation with offset



$$y = af + c$$

How to convert it to matrix-vector multiplication?

$$Ax = b$$

House Features

- #Room
- Size
- #Bedroom
- Age
- Address features: Street, Alley, ...
- Size of part1, part2, part3, part4
- Floors
- #Bathrooms
- ...

Which features are dependent on others?

02

Linear Independence

Linear Independence (Algebra)

Definition

Dependent

- ❑ For at least one $\lambda \neq 0$ $0 = \lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n, \quad \lambda \in \mathbb{R}$
- ❑ A set of vectors is dependent if at least one vector in the set can be expressed as a linear weighted combination of the other vectors in that set.

Linear Independence (Algebra)

Definition

Independent

- Only when all $\lambda_i = 0$ $0 = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n, \quad \lambda \in \mathbb{R}$
- No vector in the set is a linear combination of the others (has only the trivial solution)

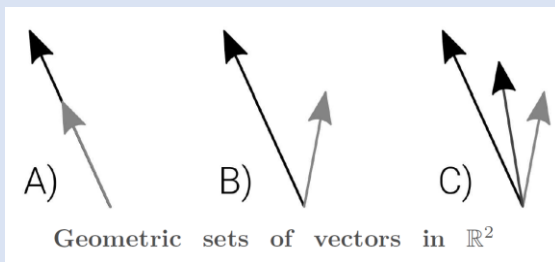
Linear Independence (Geometry)

Definition

A set of vectors is linear independent if the subspace dimensionality (its span) equals the number of vectors.

Example

□ Are these vectors linearly independent?



Example

Example

□ Let $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$.

□ a) $v_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

b) $v_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, $v_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

Properties

Theorem 1

Any set of vectors that contains the zeros vector is guaranteed to be linearly dependent.



Characterization of Linearly Dependent sets

Theorem 2

An indexed set $S = \{v_1, \dots, v_n\}$ of two or more vectors is linearly dependent **if and only if** at least one of the vectors in S is a linear combination of the others. In fact, if S is linearly dependent and $v_1 \neq 0$, then **some** v_j (with $j > 1$) is a linear combination of the preceding vectors, $v_1, \dots, v_{j-1}$.

Proof

Notes!!!

- ❑ Does not say that every vector
- ❑ Does not say that every vector in a linearly dependent set is a linear combination of the preceding vectors. A vector in a linearly dependent set may fail to be a linear combination of the other vectors.

Characterization of Linearly Dependent sets

Proof

Let j be the largest subscript for which $c_j \neq 0$. If $j = 1$, then $c_1 v_1 = 0$, which is impossible because $v_1 \neq 0$. So $j > 1$ and

$$c_1 v_1 + \cdots + c_j v_j + 0v_{j+1} + \cdots + 0v_n = 0$$

$$c_j v_j = -c_1 v_1 - \cdots - c_{j-1} v_{j-1}$$

$$v_j = \left(-\frac{c_1}{c_j}\right) v_1 + \cdots + \left(-\frac{c_{j-1}}{c_j}\right) v_{j-1}$$

Characterization of Linearly Dependent sets

Proof

If some v_j in S equals a linear combination of the other vectors, then v_j can be subtracted from both sides of the equation, Producing a linear dependence relation with a nonzero weight (-1) on v_j . [For instance, if $v_1 = c_2v_2 + c_3v_3$, then $0 = (-1)v_1 + c_2v_2 + c_3v_3 + 0v_4 + \dots + 0v_n$.] Thus S is linearly dependent.

Conversely, suppose S is linearly dependent. If v_1 is zero, then it is a (trivial) linear combination of the other vectors in S . Otherwise, $v_1 \neq 0$, and there exist weights $c_1, \dots, c_n$ not all zero, such that

$$c_1v_1 + c_2v_2 + \dots + c_nv_n = 0$$

Properties

- The vectors coming from the vector form of the solution of a matrix equation $Ax = 0$ are linearly independent

Example

- Vectors related to x_2 and x_3 are linear independent.
- Columns of A related to x_2 and x_3 are linear dependent.
- $\text{Span}\{A_1, A_2, A_3\} = \text{Span}\{A_1\}$

$$A = \begin{bmatrix} 1 & -1 & 2 \\ -2 & 2 & -4 \end{bmatrix}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

Properties

Important

- ❑ If a collection of vectors is linearly dependent, then any **superset** of it is linearly dependent.
- ❑ Any nonempty **subset** of a linearly independent collection of vectors is linearly independent.

Properties

Theorem 3

- Any set of $p > n$ vectors in $\mathbb{R}^n$ is necessarily dependent.
- Any set of $p \leq n$ vectors in $\mathbb{R}^n$ could be linearly independent.

Proof

$$\begin{matrix} & & p \\ n & \begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix} \end{matrix}$$

Exercise

Example

a. $\begin{bmatrix} 1 \\ 7 \\ 6 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 9 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 4 \\ 1 \\ 8 \end{bmatrix}$

b. $\begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 8 \end{bmatrix}$

c. $\begin{bmatrix} -2 \\ 4 \\ 6 \\ 10 \end{bmatrix}, \begin{bmatrix} 3 \\ -6 \\ -9 \\ 15 \end{bmatrix}$

Linear Dependent Properties

- Suppose vectors $v_1, \dots, v_n$ are linearly dependent:

$$c_1 v_1 + c_2 v_2 + \dots + c_n v_n = 0$$

with $c_1 \neq 0$. Then:

$$\text{span}\{v_1, \dots, v_n\} = \text{span}\{v_2, \dots, v_n\}$$

- When we write a vector space as the space of a list of vectors, we would like that list to be as short as possible. This can be achieved by iterating.

Linear combinations of linearly independent vectors

Theorem 4

Suppose x is linear combination of linearly independent vectors

$v_1, \dots, v_n$:

$$x = \beta_1 v_1 + \dots + \beta_n v_n$$

The coefficients $\beta_1, \dots, \beta_n$ are unique.

Proof

Conclusion

- 
- ❑ Step 1: Count the number of vectors (call that number p) in the set and compare to n in $\mathbb{R}^n$. As mentioned earlier, if $p > n$, then the set is necessarily dependent. If $p \leq n$ then you have to move on to step 2.
 - ❑ Step 2: Check for a vector of all zeros. Any set that contains the zeros vector is a dependent set.
 - ❑ The rank of a matrix is the estimate of the number of linearly independent rows or columns in a matrix.
- 
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03

Functions Linearly Independent



Functions Linearly Independent

- Let $f(t)$ and $g(t)$ be differentiable functions. Then they are called **linearly dependent** if there are nonzero constants c_1 and c_2 with

$$c_1 f(t) + c_2 g(t) = 0$$

for all t . Otherwise they are called **linearly independent**.

Example

Linearly dependent or independent?

- Functions $f(t) = 2 \sin^2 t$ and $g(t) = 1 - \cos^2 t$

- Functions $\{\sin^2 x, \cos^2 x, \cos(2x)\} \subset \mathcal{F}$

04

Polynomials Linearly Independent



Vector Space of Polynomials

Example

Are $(1 - x), (1 + x), x^2$ linearly independent?



Resources

- ❑ Page 97 LINEAR ALGEBRA: Theory, Intuition, Code
- ❑ Page 213: David Cherney,
- ❑ Page 54: Linear Algebra and Optimization for Machine Learning

